

RAGHAV AWASTHI

+44 7448 420483 | raghavawasthi@me.com | linkedin.com/in/raghavawasthi2005 | github.com/raghav2005

EDUCATION

University College London (UCL)

Bachelor of Science in Computer Science, First Class Honours

London, UK

Sep 2023 – Jun 2026

Dubai College

A*A*A*A*A in A-Levels and 11 grade 9s in GCSEs

Dubai, UAE

Aug 2016 – Jun 2023

EXPERIENCE

Cybersecurity Analyst

Jun 2025 – Aug 2025

Barclays

Knutsford, UK

- Owned a 60-page post-quantum cryptography/HSM assessment across 30+ services; built a capability and gap matrix, defined a five-phase roadmap, and projected a 20–30% faster migration.
- Built production Python services integrating CertNow and ServiceFirst to enrich certificate chains, classify root causes, standardise ServiceNow records, and score risk; cut triage time 25% and flagged 30+ high-risk certificates.
- Identified 100+ high-signal threat keywords and deployed detection logic to production, increasing detections 15–20% in pilot while supporting operational troubleshooting.

Full-Stack Developer

Aug 2024 – Oct 2024

Adamas Data

London, UK

- Designed and shipped a production DBMS application end-to-end using FastAPI microservices, a PostgreSQL/PostGIS data layer, and a responsive server-rendered Next.js interface.
- Built a versioned REST API with 30+ endpoints; indexed queries over 140M+ rows to achieve median latency below 3s across 150+ filters, then shipped through Dockerised CI/CD with rollback in under 10 minutes.

AI/ML and Quantum Security Intern

Jun 2024 – Aug 2024

Tech Mahindra

Pune, India

- Led a five-person R&D team to build ThaiLLM, fine-tuning a 7B-parameter model with LoRA on 120k curated records; reduced mean GPU inference latency 20% and raised domain F1 15% on a 6k-question benchmark.
- Embedded retrieval-augmented generation into Garuda using hybrid BM25 and a MiniLM re-ranker, improving top-1 retrieval MRR 25% and reducing hallucination rate to 7%.
- Researched NIST post-quantum KEM finalists and prototyped hybrid TLS 1.3 with ML-KEM in Python.

PROJECTS

Walrus-Cache-CDN (UCL Dissertation) | Rust, RocksDB, Tokio, Sui, Prometheus

Jun 2025 – Present

- Designed a lifecycle-aware cache/CDN edge layer for Walrus decentralised storage, combining an in-memory hot tier, persistent RocksDB cache, WTinyLFU/LRU admission, Sui event-driven invalidation, and operational metrics; now working with Mysten Labs toward integration into the official Walrus distribution.
- Built a reproducible 100k-request synthetic benchmark: achieved an 81.7% hit rate, reduced average retrieval latency from 85.0ms to 20.5ms, and increased throughput 4.2× over uncached retrieval.

Quantum Proximity Gateway (IBM & UCL) | Python, Rust, C++, Next.js

Oct 2024 – May 2025

- Built a distributed IoT edge-to-cloud system spanning ESP32 BLE beacons, Raspberry Pi facial recognition, and Pico USB-HID emulation to authenticate hot-desk PCs automatically as users approach.

Playo (Audiobook Generator) | Python, Firebase, OpenAI, ElevenLabs, TypeScript

Jan 2024

- Built a mobile application that turns user prompts and preferences into fully customised stories and generated audiobooks.

Assembly Code Visualiser | Node.js, Express.js, SQL, Docker

Jun 2022 – Mar 2023

- Developed a web-based assembly IDE and interactive Little Man Computer simulator for stepping through CPU and RAM state, with challenge authoring for teachers and learners.

TECHNICAL SKILLS

Languages: Python, C/C++, Rust, Java, JavaScript/TypeScript, SQL, Bash, Haskell, HTML/CSS

Backend/Web: FastAPI, Flask, Node.js, Express.js, React/Next.js, Pydantic, REST APIs, microservices

Data/ML: PostgreSQL, PostGIS, SQLite, Redis, SQLAlchemy, Pandas, NumPy, PyTorch, TensorFlow

Tools/Testing: Git, Docker, Linux, CI/CD, VS Code, Emacs, JUnit, Selenium, OpenSSL, IBM Qiskit

OPEN SOURCE, HONOURS & LEADERSHIP

- Open source:** authored merged pull requests to mactop, Docusaurus, and Microsoft Terminal, delivering a keyboard-driven Go help view and improving documentation consistency and navigation.
- Ranked 1st at school in the 2022 ISMTF mathematics competition and top five nationally in the University of Wollongong Dubai Mathalon; earned two UKMT Senior and two UKMT Intermediate Gold awards.
- Received Dubai College's Highest Achiever academic excellence award for Computer Science, 2023.
- UCL cricket vice-captain and first-year programming tutor; UCL Devs technical writer; former Dubai College 1st XI vice-captain, CS teaching assistant, and Head of Dubai Computing Community; captained my club's U18 men's volleyball team and coached its U15 women's and mixed teams.